

Effects of Recreational Marijuana Legalization
on Alcohol Consumption in California

1 Introduction

Californians, and many Americans alike, have significantly changed their opinion on marijuana in the past decade. This drug-like substance was viewed by many as dangerous to a certain extent, with multiple laws and regulations limiting it to medical use only (LaCity, 2025). In the 2000s and early 2010s, acquisition of cannabis often required a strict approval process and a doctor's note citing the need for specific treatment purposes for an illness. Commonly, the general population also perceived it as much less common and more harmful than tobacco or e-cigarettes. However, after the legalization of recreational marijuana in 2018, the California Health Interview Survey indicated that 33% of young adults consume some kind of substance, signaling a significant change in the perception of marijuana. To display this in quantitative terms, out of the 5.1 million young adults surveyed, 1.7 million people smoked substances, where 314,000 reported smoking cigarettes, 682,000 used e-cigarettes, and a shocking 1.3 million used marijuana (respondents could report more than one substance) (UCLA Health, 2020). The large proportion of Gen-Z smoking marijuana suggests a shift in young adults' substance use toward marijuana and away from tobacco and e-cigarettes. Thus, the motivating factor behind this research is to analyze the extended societal effects of this new form of substance use.

However, most of the existing literature has focused on analyzing the effects of marijuana on health factors, such as the chemical makeup of cannabis and its effects on neurological diseases (Leinen et al., 2023). There has been little research conducted on the effects of marijuana on alcohol consumption. Meanwhile, the prevalence of binge drinking among California adults rose from 15.1% in 2015 to 17.3% in 2020, according to the California Department of Public Health (California Department of Public Health, 2023). Due to the difference in the nature of alcohol – it's drunk, not smoked – and the common belief that they are substitutes for each other, there have been few studies that examined whether the consumption of marijuana has a measurable effect on alcohol consumption. Given the prevalent use of both substances, this research would aim to answer the question: What is the effect of legalized recreational marijuana on alcohol consumption in the state of California?

We chose to focus on California due to its high recreational marijuana usage compared to the national level and its diverse demographics. It was measured in 2018, 17.7% of Californians used recreational marijuana, a much higher average than the national 10.5% (California Department of Public Health, n.d.). Furthermore, California has a vastly diverse population due to its size. There is a mix of races, genders, urban and rural residents, as well as immigrants from different parts of the world (Public Policy Institute of California, 2025). Therefore, although there are external validity concerns, treating California as the treatment group provides an accurate general overview of the response of alcohol consumption from the

legalized recreational marijuana population and could generate sharper results due to the large scale of marijuana use.

We implemented a difference-in-differences regression model design in this research, identifying the treatment as the legalization of recreational marijuana in retail stores on January 1st, 2018. We retrieved the alcohol consumption data from the National Institute on Alcohol Abuse and Alcoholism. Combining this model and the data, we compared the differences between the pre-treatment difference in alcohol consumption – between California and 18 control states – and the post-treatment difference in alcohol consumption of the same two groups. The evidence is consistent with the two key model assumptions of no anticipation and parallel trends, and the estimate is stable across robustness checks. California's alcohol consumption averaged about 3% above the control states before treatment and about 16% above them after treatment, an estimated increase of 0.306 gallons of ethanol per capita. This difference is statistically significant, and the design supports a causal interpretation. However, the limitations to this research lie in external validity, since the treatment group only consists of California data. We recognize that the results of this research could only be generalized to populations with similar geographic, cultural, and demographic parameters as California. Furthermore, we will discuss the challenges moving forward with potential future research to expand the treatment group of this study.

In California, these observed effects of recreational marijuana legalization on alcohol consumption could have vast policy implications. This is because alcohol has many far-reaching impacts on different aspects of society, both positive and negative. For instance, there were around 110,000 arrests in 2021 for driving under the influence (DUI), varying by year, according to the DMV's official report (DMV, 2025). But on a positive note, there is a recorded increase in the tax revenue from alcoholic beverages purchased by consumers, reaching around \$400 million dollars a year (California Department of Tax and Fee Administration). This helps to increase the state annual budget. Due to the negative effects on the well-being of the general public and the positive effect of increasing the annual budget in the state government – amongst many other implications – the results in this study could serve as a useful reference in the decision-making of legislators on future marijuana and alcohol retail laws.

2 Research Design

During the November elections of 2016, California passed Proposition 64, which legalized “specified personal use and cultivation of marijuana for adults 21 years of age or older” (California Courts, 2016). The “specified use” clause refers to the possession, sharing, and smoking of marijuana in people 21 years of age or older. Furthermore, this proposition allows merchants to sell marijuana with an obtained state

license. However, commercially, the retail sale of recreational marijuana did not take effect until January 1st, 2018 (Paschall et al., 2021).

In this research, I aim to measure the effects of legalization of recreational marijuana on alcohol consumption by using a difference-in-differences (DiD) design. This method compares the differences of alcohol intake between the treated group and the control group from before the introduction of treatment to after the introduction of treatment. The treatment in this design will be the legalization of recreational marijuana in retail stores. The control group will be the population of states that did not legalize recreational marijuana during the complete sample period of 2001-2022. The treatment group will be the California population during the same period, but experiencing the introduction of legalized recreational marijuana beginning on January 1st, 2018. Thus, this study will use a difference-in-differences approach to analyze the effects on alcohol intake that California experienced due to the legalization of recreational marijuana, compared to other states that did not legalize through the entire sample period.

2.1 Difference-in-Differences Assumptions

To begin, there are two key assumptions given by the difference-in-differences design. First, there should be no anticipation. In other words, alcohol consumption behavior before legalization should not be affected by future recreational marijuana legalization laws or policies. Quantitatively, the first assumption can be represented as

$$Y_{st}(0) = Y_{st}(1) \text{ for } t < 2018$$

If this assumption is violated, then there is no clear start to the treatment in 2018. This means that the baseline of this model will be distorted against post-treatment changes, seriously damaging the credibility and the interpretation of the results. This assumption deserves attention here because Proposition 64 made personal possession legal in November 2016, more than a year before retail sales began. The year 2017 is therefore partly exposed to legalization, and we test for an early response in Section 4.1. The second assumption is the parallel trend assumption. In this study, it means that in the absence of recreational marijuana legalization, California's alcohol consumption would have followed the same trend as the control. In other words, the counterfactual trend of California's alcohol consumption would follow the same mean difference from the control group post-treatment as before the treatment. Quantitatively, it can be written as

$$E[Y_{s,t+1}(0) - Y_{s,t}(0) | D_s = 1] = E[Y_{s,t+1}(0) - Y_{s,t}(0) | D_s = 0]$$

If this assumption is violated, then we cannot rule out that differences the treatment group exhibits after treatment are due to factors other than legalization. This could produce invalid model results since the

treated group could have already deviated from the parallel trend before treatment. Therefore, both assumptions must be met in order for the difference-in-differences design to show a credible change in the difference between the before and after of the treatment. Neither assumption can be proven directly, because the counterfactual is never observed; Section 4 instead checks whether the data are consistent with them.

To be precise, this design aims to measure the difference between the observed outcome for alcohol consumption in California after 2018 and the counterfactual alcohol consumption that theoretically would have been observed in California after 2018 in the absence of legalized recreational marijuana. This is also known as the average treatment effect on the treated, or ATT (β). This can be formally defined as

$$\beta = E[Y_{CA,post}(1) - Y_{CA,post}(0)]$$

We can estimate β using an estimator $\hat{\beta}$. Under the two model assumptions mentioned above, the counterfactual trend is estimated by using the pre-trend differences between California and the control states to create a counterfactual trend for the post-treatment treated group. The estimator can be formally written as

$$\hat{\beta} = (\bar{Y}_{CA,post} - \bar{Y}_{control,post}) - (\bar{Y}_{CA,pre} - \bar{Y}_{control,pre})$$

This estimator equation isolates the effects of the 2018 recreational marijuana policy under the parallel trend assumption.

2.2 Difference-in-differences Regression Model

Now, in regression format, the difference-in-differences model is represented as

$$Y_{st} = \alpha + \beta (CA_s * Post_t) + \gamma_s + \delta_t + \epsilon_{st}$$

This model estimates the effect on alcohol consumption while holding constant differences between states and common shocks across years. To break it down, Y_{st} is the measurable alcohol intake, in units of ethanol consumption in gallons per capita. This will be the response variable that is estimated. α is the baseline level of alcohol intake across all data. β is the treatment effect, or the causal effect of the legalization of marijuana on alcohol intake, which only takes effect if it is the post-treatment period in California. Specifically, CA_s is a binary variable, which equals 1 if the state is within the treatment group (California) and equals 0 if the state is outside of the treatment group (non-California states), and $Post_t$ equals 1 for post-treatment years and equals 0 for pre-treatment years. Therefore, the treatment effect β only influences the response variable Y_{st} during the treatment period from 2018 to 2022 in the state of

California. This provides an estimate of the ATT, as discussed previously. Lastly, ϵ_{st} is the error term of the regression model (CA $\square$ and Post $\square$ do not enter separately, because the state and year fixed effects absorb them)

Within this model, there are also two other terms: γ_s , the unit fixed effect, and δ_t , the time fixed effect. The two fixed effect variables successfully account for two broad classes of confounders. Specifically, γ_s is the state unit fixed effect that controls for time-invariant confounders between states. For instance, it accounts for different drinking cultures in different states. Then, δ_t is the time fixed effect. This accounts for shocks that are time-dependent, such as COVID-19 and national recessions that have occurred within a specific time in the sample period. By accounting for these confounders, we are able to solidify the proposed difference-in-differences model by isolating the treatment effect. However, the fixed effects do not remove shocks that affected California alone in the same years as legalization, such as a state-specific tax or policy change. Such shocks remain the main threat to identification.

3 Data

There is one main dataset, which is retrieved from the National Institute on Alcohol Abuse and Alcoholism – NIAAA for short – *Surveillance Report #120*, published in April 2023. The data in this report contains per capita alcohol consumption on the national level and the state level that are recorded annually, collected by the Alcohol Epidemiologic Data System – or AEDS for short.

The alcohol consumption data is obtained using sales data rather than production or shipment data, since retail store sales data is a more accurate representation of consumption. Specifically, the sales data is retrieved in the form of volume tax data and converted to gallons of ethanol by AEDS. For gallons of alcohol, the institute assigned coefficients to different types of alcohol in an effort to ensure all alcohols are standardized according to their ethanol content. Specifically, the institute used the coefficient of 0.045 for beer content, 0.129 for wine content, and 0.411 for spirits content in order to convert gallons of general alcohol into gallons of ethanol. The population data was obtained from the Centers for Disease Control's WONDER online query system (National Institute on Alcohol Abuse and Alcoholism). The population age range spans 14 years and older, in order to account for underage drinking.

The summation of the three ethanol-converted values divided by the state population of the respective year results in the reported annual data value, with the units being ethanol consumption in gallons per capita. The ethanol content standard used by AEDS makes consumption comparable across beverage

types and states. Because the measure is based on sales, it also counts purchases by tourists and cross-border shoppers, which adds some measurement error.

3.1 Data Selection

From the report, we extracted the alcohol consumption data from 19 states, with each state having an annual data point within the sample period of 2001 to 2022. Out of the 19 states, California is the only state within the treatment group, having its annual measurements of alcohol consumption as the treatment data. The trend plots use the annual average of the control states as the control group, while the regressions use each control state's annual data separately. The 18 control states that have not legalized recreational marijuana during the entire sample period are Alabama, Arkansas, Georgia, Florida, Idaho, Indiana, Iowa, Kansas, Kentucky, Louisiana, Mississippi, Nebraska, North Carolina, South Carolina, Tennessee, Texas, Wisconsin, and Wyoming. No differentiation was made between control states that have complete bans on marijuana and those that have legalized prescription marijuana. This is because prescription marijuana is relatively hard to obtain in states that utilize it. For instance, in the state of Florida, which is a control state, prescription marijuana requires a doctor's note, a registry card with Florida's Medical Marijuana Registry, and an online application charge of 75 dollars (Medical Marijuana Treatment Clinics of Florida). In comparison, other control states vary in the strictness of prescription marijuana laws, but the overall pattern suggests the general presence of a rigorous legal process to obtain marijuana for medical purposes. We therefore expect differences in medical marijuana policy across control states to have only a small effect on alcohol consumption, although we do not test this directly. This research also does not measure illegal marijuana use. It assumes that illegal access did not change differently in California and the control states over the sample period. State-level survey estimates, such as those from the National Survey on Drug Use and Health, could be used to relax this assumption in future work.

Moreover, the annual data measured by each state occur from the first annual date to the last annual date. For instance, the 2018 California data represents per capita consumption of ethanol in gallons from January 1st, 2018 to December 31st, 2018. Since legalization of recreational marijuana took effect in the marketplace on January 1st, 2018, in California, the 2018 California sample effectively recognizes the first-year effect of treatment. In turn, the 2017 annual data would serve as the baseline year ($t = -1$), which is the last period before the treatment occurs.

3.2 Summary Statistics

A summary statistics table was created to compare the two groups over the full sample period of 2001 to 2022. As shown, California has a 0.14 higher average mean of ethanol consumption in gallons per capita than the control states, with a lower standard deviation by 0.13. California's lower standard deviation partly reflects that it is a single state, while the control group's standard deviation also captures differences among 18 states. Over the pre-treatment period alone (2001 to 2017), California averaged 2.28 gallons per capita compared with 2.21 for the control states. A difference-in-differences design does not require the two groups to have equal levels, only parallel trends. To assess the two difference-in-differences model assumptions, we will conduct more detailed data analysis in the results section of this research. Another important note is that the number of observations is much higher for the control group than for the treatment group, since the control group contains 18 states observed over 22 years (396 state-years), compared with 22 years for California. Nevertheless, there are limitations to the design's external validity and precision, which will be further discussed in the conclusion section of this paper.

Table 1: Summary Statistics: Alcohol Consumption by Group (2001-2022)

	California			Control States		
	Mean	SD	N	Mean	SD	N
y	2.37	0.20	22	2.23	0.33	396

4 Results

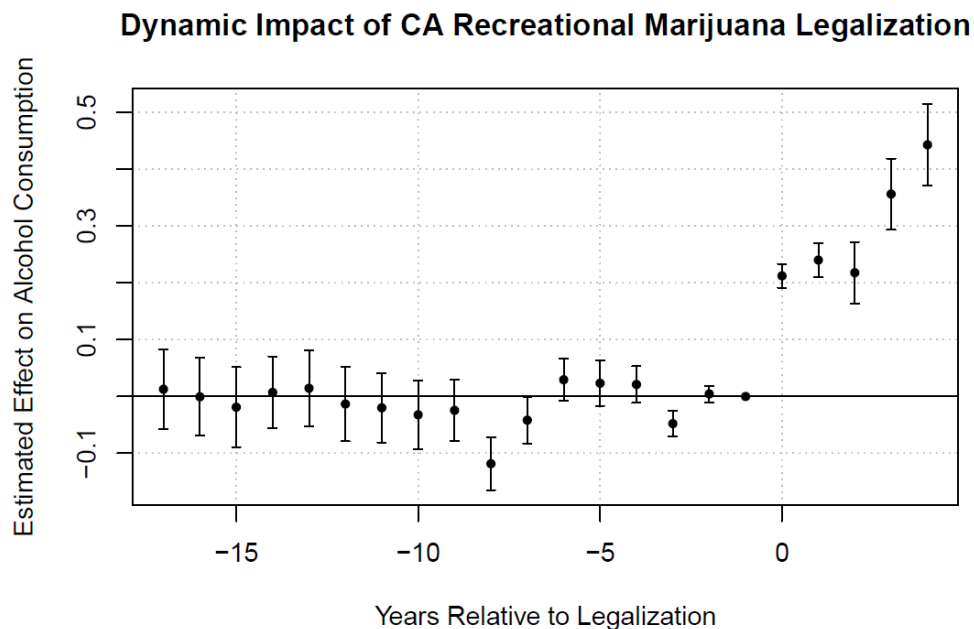
In this section, we will examine the quantitative effects of legalized recreational marijuana on alcohol consumption by comparing the data in California and in the control states. The results indicate an average effect of 0.306 gallons per capita consumption of ethanol. Before treatment, California consumed on average 3.0% more alcohol than the control states, a gap of 0.066 gallons per capita. After treatment, the gap averaged 16.1% (0.372 gallons), and by 2022 California's consumption was 22.1% higher than the control states (0.521 gallons). The validity of the difference-in-differences regression model will also be analyzed, assessing the no anticipation and parallel trend assumptions. A robustness check will be implemented to ensure that no single state or time period drives the main findings.

4.1 No Anticipation Assumption

The first assumption of the difference-in-differences model is no anticipation, which means that the two groups' pre-treatment effects should be roughly equal. This would indicate that the treatment in the

post-2018 period does not have an effect on the pre-2018 period. This will be assessed using an event study plot, which serves as a placebo test, with the goal of demonstrating no observable effects on alcohol consumption before the legalization of recreational marijuana.

In the event study plot, the y-axis plots the estimated difference in alcohol consumption relative to the baseline year ($t = -1$ in 2017) and relative to the control states (average of 18 control states), after controlling for the state and year fixed effects. As displayed, years 2001 to 2016 demonstrated roughly no difference in estimated effects compared to the baseline year 2017, with each confidence interval overlapping with the x-axis, indicating no effects – except for two anomaly years of 2010 and 2015. The cause of these two dips is unclear. Because California's series returns to its earlier trend the following year, they look like one-year fluctuations in a single state's data rather than a change in trend. The baseline year 2017 is at $y=0$ with no confidence interval, since it is being compared to its own identity. Years 2018 to 2022 showed a dramatic increase in the estimated effect on alcohol consumption, with the difference in ethanol consumption jumping to a rough increase of 0.44 gallons in 2022 compared to that of 2017.

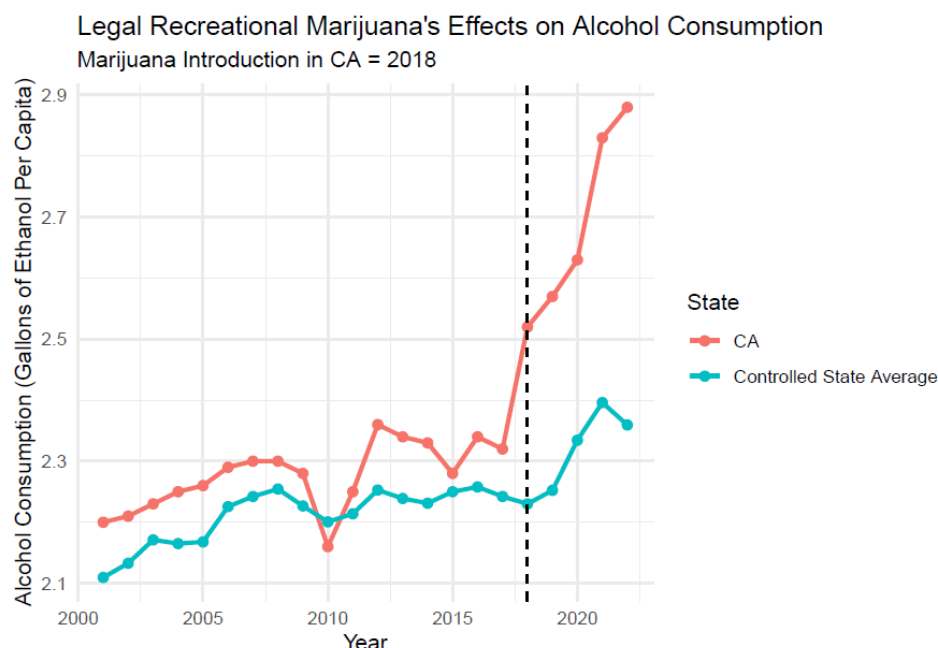


Therefore, the event study is consistent with the no anticipation assumption. According to the event study plot, there are roughly no differences in alcohol consumption in the pre-treatment period compared to the 2017 consumption data, which is the baseline statistics. Because Proposition 64 passed in November 2016, consumption could have responded in 2017, before retail sales began. If it had, the 2017 baseline would be raised and the post-treatment effects understated. However, California's 2017 gap over the

control states (0.078 gallons) is close to its 2001–2016 average, so there is little sign of this. During the post-treatment period, there is a statistically significant increase in alcohol consumption compared to the baseline year, with no overlapping intervals compared to the pre-treatment intervals. The flat pre-treatment coefficients are also the strongest evidence for the parallel trend assumption discussed next, since they show no diverging trend between California and the control states before 2018.

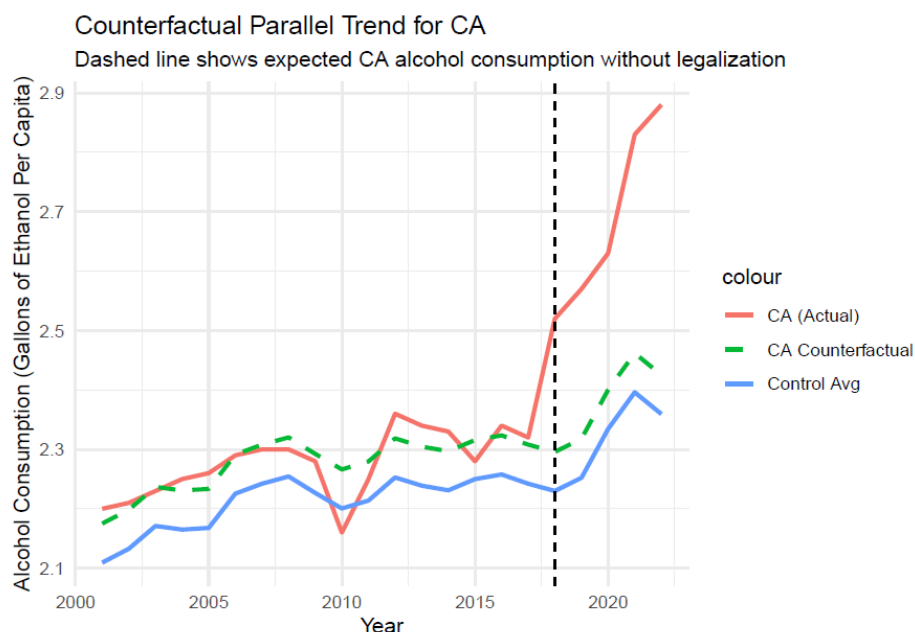
4.2 Parallel Trend Assumption

The second assumption is the parallel trend assumption, which means that the counterfactual trend should follow the same differences from the control group as in the pre-treatment period. To illustrate the presence of a parallel trend in the pre-treatment period, we simply plotted California alcohol consumption alongside the average consumption of control states. This graph displays a clear parallel trend from 2001 to 2017. The abnormal lows of consumption in 2010 and 2015 caused slight deviations in the treated trend, but the overall progression of the treatment group stayed relatively similar to the control group, with an average pre-treatment gap of about 0.07 gallons of ethanol per capita.



Having focused on and observed the parallel trend pre-treatment, we will now examine the counterfactual trend after the introduction of legal recreational marijuana in 2018. The counterfactual trend is graphed using the average difference between the treated California group and the control state group during the pre-treatment period. By applying this baseline difference to the post-treatment period, we can observe the hypothetical trend that California would have followed in the absence of treatment. Because this line is constructed by assuming parallel trends, it illustrates the estimated effect rather than testing the

assumption. There is a large difference between the counterfactual trend and the actual observed alcohol consumption in California. Under the parallel trend assumption, this difference is the estimated effect of Proposition #64, or the introduction of legalized recreational marijuana policy, on alcohol consumption in California.



4.3 Summary Results

The difference-in-differences regression shows that California in the post-treatment period experienced an average increase of 0.306 gallons of ethanol consumption per capita. This is the average causal effect of the legalization of recreational marijuana on alcohol consumption.

Table 2: Difference-in-Differences Estimates of the Effect of Recreational Marijuana Legalization on Alcohol Consumption in CA

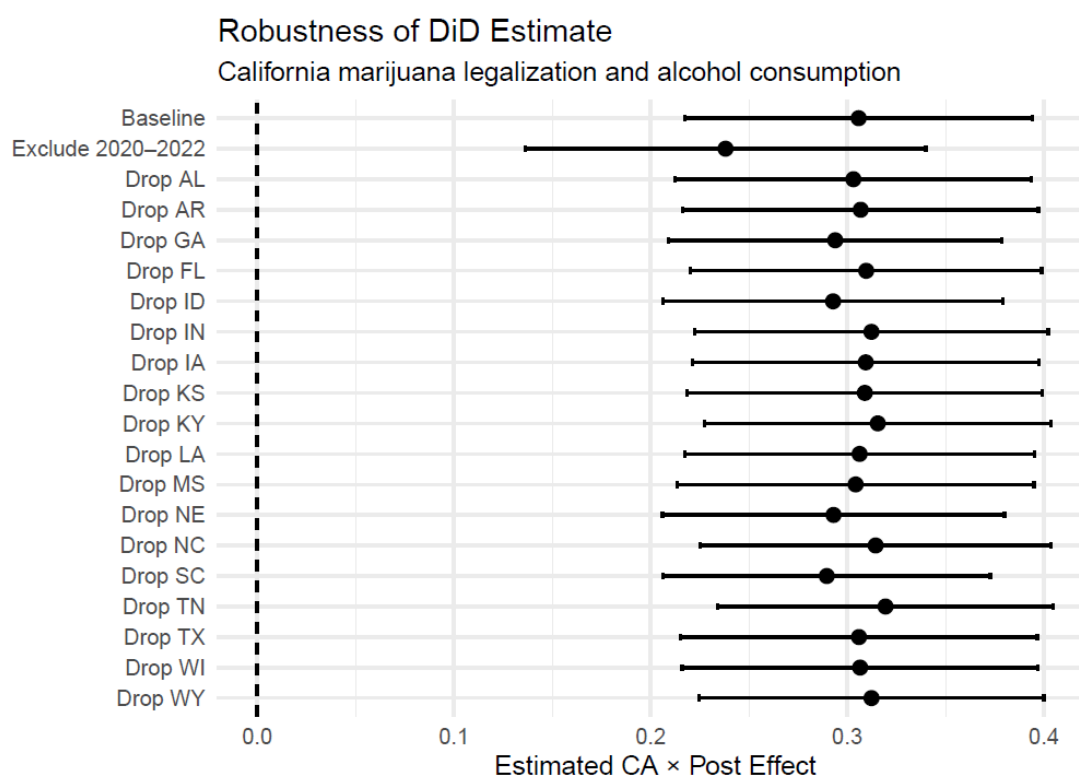
	(1)
California $\times$ Post (DiD Effect)	0.318
	(0.035)
Num.Obs.	418
R ²	0.935

Note that the standard errors in this model are clustered at the state level, and that the displayed results already account for the state and year fixed effects. The high R^2 indicates a good fit of the model. In other words, roughly 93.5% of the variation in alcohol consumption is explained by the difference-in-differences regression model. This could largely be attributed to the unit and time fixed

effects absorbing most of the variation in the regression. A high R^2 describes model fit and does not by itself show that the estimate is causal; the causal interpretation rests on the assumptions assessed in Sections 4.1 and 4.2. In addition, with only one treated state, standard errors clustered by state can understate the true uncertainty, so the reported standard error should be read with some caution.

4.3 Robustness of Difference-in-differences Estimate

Lastly, a robustness test was performed to ensure that the result that we concluded above was not influenced by one specific factor: either a period of time or a specific state. In this robustness test, we ran the regression model many times on restricted samples: once excluding the COVID period (2020–2022), and once dropping each control state in turn. For each sample, the estimated effect and its 95% confidence interval are plotted.



No single control state changes the estimate meaningfully: every estimate stays between about 0.29 and 0.32, close to the baseline of 0.306. Excluding 2020–2022 lowers the estimate to about 0.24, roughly 20% below the baseline, but its confidence interval still overlaps the baseline and excludes zero. The drop suggests part of the effect falls in the pandemic years, when California’s gap over the control states widened most (0.43 gallons in 2021 and 0.52 in 2022). Overall, we can conclude that the observed increase in alcohol consumption in California after the legalization of recreational marijuana was not driven by any single control state, and a positive effect remains even without the pandemic years. (These

intervals use conventional rather than state-clustered standard errors, so they are wider than the interval implied by Table 2.)

5 Conclusion

This research was able to estimate the causal effects of legalization of recreational marijuana on alcohol consumption. The credibility of the data source was discussed, which includes a well-defined alcohol consumption measurement standard and calculation. Using the data, we found evidence consistent with the two key assumptions of the difference-in-differences model (no anticipation and parallel trend). The event plot demonstrated the stable pre-treatment trend and the noticeable jump in alcohol consumption directly following the retail introduction of recreational marijuana in California. The trend plot showed California and the control states moving in parallel before 2018, and the counterfactual trendline illustrated the size of the estimated effect. Lastly, a robustness test was performed to verify that no single control state or time period is driving the results. Overall, we saw that the average level of alcohol consumption in California was roughly 16% higher than the control states in the post-treatment period, compared to roughly 3% higher during the pre-treatment period. This significant difference in the differences in alcohol consumption is consistent with recreational marijuana and alcohol acting as complements rather than substitutes in California, at least at the aggregate level.

The support from parallel trends, fixed effects, and the robustness check strengthens internal validity. There could be potential unobserved shocks, but overall it is reasonable to conclude that the estimated effect is causal. However, there still are limitations to this research, which include spillover effects of illegal marijuana trading that are sparked by the legalization of the use of recreational marijuana, or any unnoticed changes in tax bills or concurrent policies that influenced alcohol intake directly after 2018. Part of the estimated effect also falls in the pandemic years of 2020 to 2022, and with only one treated state, the standard errors may understate the true uncertainty. Moreover, there is a major external validity issue: the treatment group only consists of the California population. Although California is diverse, large, and spans many regions, the singularity of this data allows more fluctuations than the average of the control states, such as seen in 2010 and 2015. Therefore, generalization is only applicable to populations similar to that of California and is limited by external validity issues. For instance, smaller states with no urban cities at all may reflect a completely different alcohol effect than California did.

Therefore, there is much future research to be completed to strengthen external validity by considering other states that have also adopted recreational marijuana policies. However, one main challenge is consolidating multiple treatment states into a single regression line like the control states, due to each

state's different adoption timelines. Furthermore, different states may have differing policies for recreational marijuana as well. For instance, the state of Virginia currently allows the possession and use of recreational marijuana, but not the sale of it. Only starting January 1st, 2027, will the retail sale of recreational marijuana be legalized (Virginia's home for Public Media, 2026). The differences in marijuana policies pose data consolidation challenges amongst treatment states; thus, this is an important facet to consider when analyzing multiple states in the treatment group for future research. Lastly, since recreational marijuana was introduced in California for less than ten years, there would be interest in continuing to observe the California alcohol consumption trend for a longer duration of the sample period to observe any deviation in the post-treatment trends.

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